

Spring

The Spring Self-Evolving Potential Function Trainer:

A Training-Is-Audit Framework for
Molecular Dynamics Potential Learning

SCX

202671

v1.0 — — SCX —

Abstract

SPRING—— NEPDeepMDACE SPRING $M > 1$ YAJIE Lyapunov Ψ_t M_t CERCIS
Spring $\Psi_t \rightarrow 0$ $t \rightarrow \infty$ $1 M_t \Sigma_0(\mathcal{D})$ -2 YAJIE M_t 3 SPRINGNEP $M = 1$ DeepMDM =
1ACEM = 1 SPRING””
YajieLyapunovCercisSpring

Contents

1

1.1

Molecular Dynamics, MDV($\mathbf{r}_1, \dots, \mathbf{r}_N$)

Lennard-JonesEAMReaxFF

Machine Learning Potentials, MLPs Behler-Parrinello2007

- **NEP**Neuroevolution Potential, 2022 $M = 1$
- **DeepMD**Deep Potential Molecular Dynamics, 2018 $-M = 1$
- **ACE**Atomic Cluster Expansion, 2019 $M = 1$
- **GAP**Gaussian Approximation Potential, 2010 $M = 1$
- **MTP**Moment Tensor Potential, 2016 $M = 1$

———— / ———”””””

1.2

SCX [?, ?] $\mathcal{D} = \{(\mathbf{R}_i, E_i, \mathbf{F}_i)\}_{i=1}^{N_{\text{data}}} \quad \mathbf{R}_i E_i \mathbf{F}_i \quad V_\theta(\mathbf{R})$

$$\text{RMSE}_E = \sqrt{\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (V_\theta(\mathbf{R}_i) - E_i)^2}, \quad \text{RMSE}_F = \sqrt{\frac{1}{3N_{\text{test}}N_{\text{atoms}}} \sum_{i=1}^{N_{\text{test}}} \|\mathbf{F}_\theta(\mathbf{R}_i) - \mathbf{F}_i\|^2} \quad (1)$$

$\text{RMSE}_E = 1 \text{ meV/atom} \quad 1 \text{ meV/atom}$

(1) 1%50%

(2) $\mathbf{R}^*$ ————

(3) $V_{\theta_1} V_{\theta_2} \text{RMSE}$ ——

Audit Trilemma

Definition 1.1. V_θ

(A1) Error Localization $\mathbf{R} V_\theta(\mathbf{R}) \varepsilon E_{\text{ref}}(\mathbf{R})$

(A2) OOD Detection $\mathbf{R}$

(A3) Multi-Solution Distinguishability RMSE

- **NEP** $M = 1$ (A1)——SCX2 [?]
- **DeepMD** $M = 1$ -(A2)DeepMDDropout ensembleDeep Ensemble
- **ACE** $M = 1$ (A3)ACE”””””——

1.3 Spring

SPRING”””” Spring Springt

$$\boxed{\text{Spring}(t) \mapsto \left(\underbrace{V_{\text{consensus}}^{(t)}}_{\text{Yajie}}, \underbrace{\mathbf{s}_{\text{YAJIE}}^{(t)}}_{\text{Yajie}}, \underbrace{\Psi_t}_{\text{Lyapunov}}, \underbrace{M_t^{\text{auto}}}_{M_t}, \underbrace{S_{\text{CERCIS}}^{(t)}}_{\text{Cercis}} \right)} \quad (2)$$

—— ””Spring——

1.4

- (1) 2Spring
- (2) 3—— $M > 1$
- (3) 4Yajie——
- (4) 5 M_t ——
- (5) 6Cercis——
- (6) 7Lyapunov
- (7) 8NEP/DeepMD/ACE
- (8) 9——
- (9) 10

2 Spring

2.1

SpringSelf-Evolving Autonomous System

1 Multi-Expert Bootstrap Layer $\mathcal{D}M_t\mathcal{D}_1, \dots, \mathcal{D}_{M_t}$

$$I(\mathcal{D}_m; \mathcal{D}_{m'}) \leq \delta_{\text{indep}}, \quad \forall m \neq m' \quad (3)$$

$I(\cdot; \cdot)\delta_{\text{indep}}$ stratified bootstrap

2 Independent Training Layer $f_m m = 1, \dots, M_t \mathcal{D}_m m\theta_m$

$$\theta_m^{(t)} = \arg \min_{\theta} \mathcal{L}_m(\theta; \mathcal{D}_m) + \Omega(\theta) \quad (4)$$

$\mathcal{L}_m m\Omega(\theta)$ — SCXA1A2

3 YajieYajie Consensus Layer $\mathbf{R}M_t$

$$\hat{E}_m = f_m(\mathbf{R}), \quad \hat{\mathbf{F}}_m = -\nabla_{\mathbf{R}} f_m(\mathbf{R}), \quad m = 1, \dots, M_t \quad (5)$$

Yajie4 $\hat{E}_{\text{YAJIE}} s_{\text{YAJIE}}(\mathbf{R}) \in [0, 1]$

4 Self-Evolution Regulation Layer Yajie

- τ_{noise}
- Lyapunov Ψ_t
- $M_t M_{t+1}$

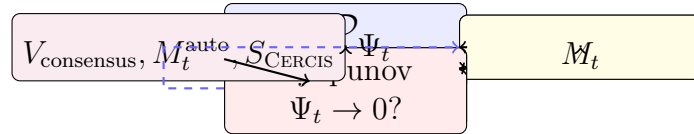


Figure 1: Spring $\rightarrow\rightarrow\rightarrow$

2.2 Spring

??Spring t — (??) —

Definition 2.1 (Spring). $\mathcal{D}tSpring$

(1) $V_{\text{consensus}}^{(t)}$

$$V_{\text{consensus}}^{(t)}(\mathbf{R}) = \sum_{m=1}^{M_t} \omega_m^{(t)} \cdot f_m^{(t)}(\mathbf{R}) \quad (6)$$

$$\omega_m^{(t)} = (\sigma_m^{(t)})^{-2} / \sum_j (\sigma_j^{(t)})^{-2} \sigma_m^{(t)} m \text{ Yajie } V_{consensus}^{(t)}$$

$$(2) \text{ \textbf{Yajies}}_{\text{Yajie}}^{(t)} \mathbf{R}_i \in \mathcal{D} \text{ Yajie} [0, 1] M_t \mathbf{s}_{\text{YAJIE}}^{(t)} = (s_1^{(t)}, \dots, s_{N_{data}}^{(t)})$$

$$(3) \text{ \textbf{Lyapunov}} \Psi_t$$

$$\Psi_t = \frac{1}{N_{data}} \sum_{i=1}^{N_{data}} \text{Var}_{m=1, \dots, M_t} [f_m^{(t)}(\mathbf{R}_i)] \tag{7}$$

$$\Psi_t \text{SpringLyapunov}$$

$$(4) M_t^{auto}$$

$$M_t^{auto} = \Xi(\mathcal{H}(\mathcal{D}); \Psi_t, \mathbf{s}_{\text{YAJIE}}^{(t)}) \tag{8}$$

$$\mathcal{H}(\mathcal{D}) \text{SHA-256} \Xi 5 M_t^{auto} \text{---}$$

$$(5) \text{ \textbf{Cercis}} S_{\text{Cercis}}^{(t)}$$

$$S_{\text{CERCIS}}^{(t)} = Q^{(t)} + \eta \cdot N^{(t)} \tag{9}$$

$$Q^{(t)} \text{ YajieLyapunov } N^{(t)} \eta = 0.2$$

2.3

??Spring

Table 1: Spring										
	NEP	DeepMD ACE		GAP	Spring					
M	$M = 1$	$M = 1$	$M = 1$	$M = 1$	$M_t > 1$					
=										
					Yajie					
					Lyapunov					
	RMSE	RMSE	RMSE		Cercis					
M										
ensemble										

3

3.1 $M = 1M > 1$

”” V^*

$E_0(\mathbf{R})$ Schrödinger $V(\mathbf{R})$ ——
SCX1 [?]

Theorem 3.1 (— 1). $\mathcal{DN}_{data}iE_i\mathbf{F}_i\mathcal{F}^{(M_t)} = \{f_1, \dots, f_{M_t}\}M_t$ (i) (??)(ii) $f_m\mathbf{R}M_t\Delta$

$$\mathbb{P}\left(\bigcap_{m=1}^{M_t}\{|f_m(\mathbf{R}) - E_{ref}(\mathbf{R})| > \Delta\}\right) \leq \exp\left(-2M_t^{eff} \cdot \frac{\Delta^2}{\bar{\sigma}^2}\right) \tag{10}$$

$$M_t^{eff} = M_t/(1 + \bar{\rho}_t) \ \bar{\rho}_t \ \bar{\sigma}^2$$

Proof. $Z_m = \mathbb{I}[|f_m(\mathbf{R}) - E_{ref}(\mathbf{R})| > \Delta]$ (i) $Z_m \ \bar{\rho}_t \ \mathbf{R}\Delta p_\Delta = \mathbb{P}(Z_m = 1)$
 $Z_m\mathbf{R}$ Hoeffding M_t^{eff}

$$\mathbb{P}\left(\sum_{m=1}^{M_t} Z_m = 0\right) = \prod_{m=1}^{M_t} \mathbb{P}(Z_m = 0 \mid \mathbf{R}) \cdot (1 + \mathcal{O}(\bar{\rho}_t)) \tag{11}$$

$$\leq (1 - p_\Delta)^{M_t^{eff}} \tag{12}$$

$$\leq \exp\left(-M_t^{eff} \cdot p_\Delta\right) \tag{13}$$

$$\text{Chernoff}p_\Delta \geq 2\Delta^2/\bar{\sigma}^2\text{(ii) (??)} \qquad \qquad \qquad \square \qquad \qquad \qquad \square$$

Corollary 3.2. $\varepsilon\mathbb{P}() \leq \varepsilon$

$$M_{\min}(\varepsilon, \Delta) = \left\lceil \frac{\bar{\sigma}^2 \cdot \ln(1/\varepsilon)}{2\Delta^2} \cdot (1 + \bar{\rho}) \right\rceil \tag{14}$$

$$\bar{\sigma} \approx 5 \text{ meV/atom} \ \Delta = 2 \text{ meV/atom} \ \varepsilon = 0.0199\% \ \bar{\rho} = 0.3 \ M_{\min} \approx 7.2 \rightarrow M_{\min} = 8$$

3.2

””ensemble learning baggingboosting Spring

Remark 3.3 (vs.). $M\bar{f}(\mathbf{R}) = \frac{1}{M} \sum_m f_m(\mathbf{R})$ M —— Spring M_t Yajie M_t ——5

Proposition 3.4. $M > 1$ *Spring*

$$\Delta I^{(t)} = I(V_{consensus}^{(t)}; V^* \mid \mathcal{D}) - I(V_{consensus}^{(t-1)}; V^* \mid \mathcal{D}) \geq 0 \tag{15}$$

$$\Psi_t \geq \Psi_{t-1}$$

$$\cdot \ MV^*M \text{ Yajie } I(; V^*) \geq I(f_m; V^*)m$$

$$\text{Lyapunov}\Psi_t7 \qquad \qquad \qquad \square \qquad \qquad \qquad \square$$

4 Yajie

4.1 Yajie

YAJIE [?]

Spring $\mathbf{R} M_t \{\hat{E}_m(\mathbf{R})\}_{m=1}^{M_t}$ Yajies $s_{\text{YAJIE}}(\mathbf{R})$

$$s_{\text{YAJIE}}(\mathbf{R}) = \exp \left(-\frac{1}{M_t} \sum_{m=1}^{M_t} \left(\frac{\hat{E}_m(\mathbf{R}) - \hat{E}_{\text{YAJIE}}(\mathbf{R})}{\sigma_{\text{calib},m}} \right)^2 \right) \quad (16)$$

- $\hat{E}_{\text{YAJIE}}(\mathbf{R}) = \sum_m \omega_m \hat{E}_m(\mathbf{R})$ $\omega_m = \sigma_m^{-2} / \sum_j \sigma_j^{-2}$
- $\sigma_{\text{calib},m} m \text{Yajie} \mathcal{D}_{\text{calib}}$
- $s_{\text{YAJIE}} \in [0, 1]$ $s_{\text{YAJIE}} = 1$ $s_{\text{YAJIE}} \rightarrow 0$

4.2

Theorem 4.1 (Yajie — 2). $\mathbf{R} E_{\text{ref}}(\mathbf{R}) E_{\text{obs}}(\mathbf{R}) = E_{\text{true}}(\mathbf{R}) + \epsilon$ $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ $M_t \text{Yajies}_{\text{YAJIE}}(\mathbf{R})$

$$\mathbb{E}[s_{\text{YAJIE}}(\mathbf{R})] = \left(1 + \frac{\sigma_\epsilon^2}{\bar{\sigma}_{\text{expert}}^2} \right)^{-M_t/2} \quad (17)$$

$$\bar{\sigma}_{\text{expert}}^2 = \frac{1}{M_t} \sum_m \sigma_{\text{calib},m}^2$$

Proof. m

$$\hat{E}_m(\mathbf{R}) = E_{\text{true}}(\mathbf{R}) + \epsilon + \delta_m$$

$$\delta_m \sim \mathcal{N}(0, \sigma_m^2)$$
 $\epsilon \perp \delta_m$

$$\hat{E}_{\text{YAJIE}}(\mathbf{R}) = E_{\text{true}}(\mathbf{R}) + \epsilon + \bar{\delta}$$
 $\bar{\delta} = \sum_m \omega_m \delta_m$ $\hat{E}_m - \hat{E}_{\text{YAJIE}} = \delta_m - \bar{\delta}$ (??)

$$\mathbb{E} \left[\left(\frac{\delta_m - \bar{\delta}}{\sigma_{\text{calib},m}} \right)^2 \right] = 1 - \frac{2}{M_t} + \mathcal{O}(M_t^{-2}) \approx 1$$

$$\sigma_{\text{calib},m}^2 \approx \text{Var}(\delta_m - \bar{\delta})$$

$$\sigma_\epsilon = 0 \mathbb{E}[s_{\text{YAJIE}}] \approx \exp(-1/2)^{M_t} \hat{E}_{\text{YAJIE}} \epsilon$$
 (??) $\mathbb{E}[(\hat{E}_m - \hat{E}_{\text{YAJIE}})^2] = \sigma_{\text{calib},m}^2 + \sigma_\epsilon^2 \cdot (1 - 2\omega_m + \sum_j \omega_j^2)$ $\square$ $\square$

Corollary 4.2. $M_t \text{Yajie} \sigma_\epsilon > 0$ $\sigma_\epsilon = 0$

$$\frac{\mathbb{E}[s_{\text{YAJIE}} \mid \sigma_\epsilon > 0]}{\mathbb{E}[s_{\text{YAJIE}} \mid \sigma_\epsilon = 0]} = \left(1 + \frac{\sigma_\epsilon^2}{\bar{\sigma}_{\text{expert}}^2} \right)^{-M_t/2} \quad (18)$$

$$M_t M_t = 8\sigma_\epsilon = 2\bar{\sigma}_{\text{expert}} 5^{-4} = 0.0016 \text{---}$$

Spring YAJIE

4.3

SpringYajie

Protocol 4.3 (Yajie). t

1 $N_{\text{data}}\{s_i^{(t)}\}_{i=1}^{N_{\text{data}}}$

2 $\tau_{\text{noise}}^{(t)}$

$$\tau_{\text{noise}}^{(t)} = Q_{p_{\text{noise}}}(\{s_i^{(t)}\}) \quad (19)$$

$$Q_{pp_{\text{noise}}} \in [0.05, 0.20]$$

3 $\tau_{\text{noise}}^{(t)}$

$$\mathcal{D}^{(t+1)} = \{\mathbf{R}_i \in \mathcal{D} : s_i^{(t)} \geq \tau_{\text{noise}}^{(t)}\} \quad (20)$$

4 $\mathcal{H}_{\text{noise}}^{(t)}$

Remark 4.4. Spring — $M_t M_t - 11$ p_{noise} — Spring Yajie

5 M_t

5.1

Spring M_t M_t — M_t

Definition 5.1 (M_t^{auto}). $\mathcal{D}SHA\text{-}256\mathcal{H}(\mathcal{D}) = SHA\text{-}256(\mathcal{D})$ t

$$\boxed{M_t^{\text{auto}} = \Xi(\mathcal{H}(\mathcal{D}); \Psi_{t-1}, \mathbf{s}_{\text{YAJIE}}^{(t-1)}, \Sigma_0(\mathcal{D}))} \quad (21)$$

- $\mathcal{H}(\mathcal{D})$ — M_t
- Ψ_{t-1} *Lyapunov* —
- $\mathbf{s}_{\text{YAJIE}}^{(t-1)}$ *Yajie* —
- $\Sigma_0(\mathcal{D})??$
- $\Xi : \{0, 1\}^{256} \times \mathbb{R}^+ \times [0, 1]^{N_{\text{data}}} \times \mathbb{R}^+ \rightarrow \mathbb{N}_{>1}$

5.2 $\Sigma_0(\mathcal{D})$

Definition 5.2. $\mathcal{D}[E_{\min}, E_{\max}]$

$$H_E(\mathcal{D}) = - \int_{E_{\min}}^{E_{\max}} p(E) \log p(E) dE \quad (22)$$

$p(E)$ $\Sigma_0(\mathcal{D})$

$$\Sigma_0(\mathcal{D}) = \frac{H_E(\mathcal{D})}{H_{\max}} \cdot \left(1 + \frac{\sigma_E}{\bar{E}}\right) \cdot \left(1 + \frac{N_{\text{species}}}{N_{\text{species}}^{\text{ref}}}\right) \quad (23)$$

$$H_{\max} = \log(E_{\max} - E_{\min}) \sigma_E \bar{E} N_{\text{species}} N_{\text{species}}^{\text{ref}} = 1 \quad \Sigma_0(\mathcal{D}) \in [0, \infty)$$

Theorem 5.3 ($M_t - 3$). *Spring* $M_t \Sigma_0(\mathcal{D})$

$$\Sigma_0(\mathcal{D}_1) \geq \Sigma_0(\mathcal{D}_2) \implies M_t^{\text{auto}}(\mathcal{D}_1) \geq M_t^{\text{auto}}(\mathcal{D}_2) \quad (24)$$

M_t

$$M_{\min} \leq M_t^{\text{auto}} \leq M_{\max} \quad (25)$$

$$M_{\min} = 3SCX1M > 1 \quad M_{\max} = \min(32, \lfloor N_{\text{data}}/N_{\text{min_per_expert}} \rfloor)$$

Proof. Ξ

- (1) $\mathcal{H}(\mathcal{D})_{\text{seed}} = \text{int}(\mathcal{H}(\mathcal{D})[0 : 8], 16)$
- (2) $\alpha(\mathcal{D}) = \min(1, \Sigma_0(\mathcal{D})/\Sigma_0^{\text{ref}}) \Sigma_0^{\text{ref}} \Sigma_0$
- (3) $\beta_t = \max(0, 1 - \Psi_{t-1}/\Psi_0) \Psi_0$

$$(4) \quad \gamma_t = 1 - \frac{1}{N_{\text{data}}} \sum_i \mathbb{I}[s_i^{(t-1)} < \tau_{\text{noise}}]$$

$$(5) \quad M_t$$

$$M_t^{\text{auto}} = \lfloor M_{\min} + (M_{\max} - M_{\min}) \cdot \alpha(\mathcal{D}) \cdot \beta_t \cdot \gamma_t \rfloor$$

$$\alpha(\mathcal{D})\Sigma_0(\mathcal{D})\beta_t\gamma_t \quad M_t^{\text{auto}} \qquad \square \qquad \square$$

Remark 5.4. ””Symbiotic Binding

- $\mathcal{D}M_tM_t$
- $M_t\mathcal{H}(\mathcal{D})M_tM_t$

Spring Spring

5.3 M_t

Spring M_t —— ?? M_t

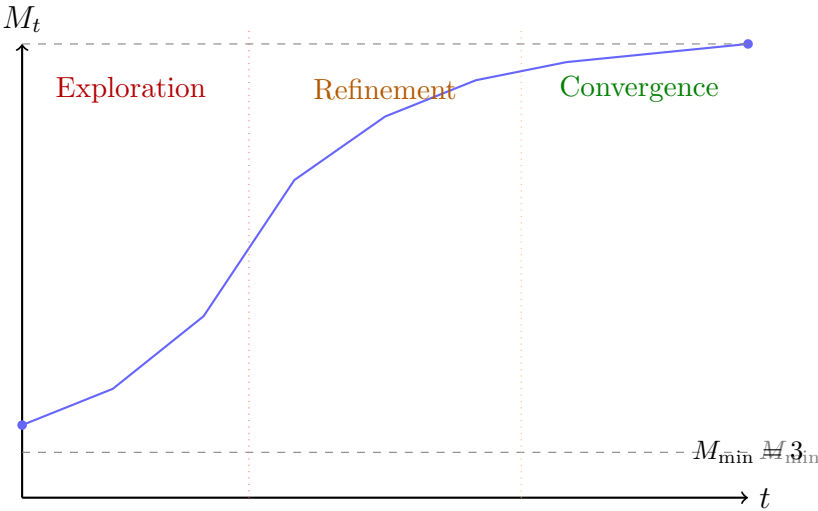


Figure 2: $M_t(1)$ $M_t(2)$ $M_t(3)$ M_t

6 Cercis

6.1

RMSE RMSE

(1) RMSE——”””””

(2) RMSERMSE

(3) NEPRMSE=1 meV/atomDeepMDRMSE=1 meV/atom NEPDeepMDRMSE——
RMSE

CERCIS [?] **PrecisionCoverage**

Definition 6.1 (CERCIS —). $SpringV_{consensus}^{(t)}$ CERCIS

$$\boxed{S_{\text{CERCIS}}(V) = Q_{\text{prec}}(V) + \eta \cdot N_{\text{cov}}(V)} \quad (26)$$

- $Q_{\text{prec}}(V) \in [0, 1]$ **Quality Guarantee**
- $N_{\text{cov}}(V) \in [0, 1]$ **Coverage**
- $\eta = 0.2$

6.2 Q_{prec}

Q_{prec} 1.2

$$Q_{\text{prec}}(V) = \frac{1}{3} (q_{\text{loc}}(V) + q_{\text{ood}}(V) + q_{\text{multi}}(V)) \quad (27)$$

(1) q_{loc}

$$q_{\text{loc}}(V) = 1 - \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbb{1}[|V_{\text{consensus}}(\mathbf{R}_i) - E_i| > \varepsilon \wedge s_{\text{YAJIE}}(\mathbf{R}_i) > \tau_{\text{conf}}]$$

Spring $q_{\text{loc}} \rightarrow 1$

(2) q_{ood}

$$q_{\text{ood}}(V) = \text{AUROC}(s_{\text{YAJIE}}(\mathbf{R}), \mathbb{1}[\mathbf{R} \notin \text{supp}(\mathcal{D})])$$

YajieOODAUROC Spring

(3) q_{multi}

$$q_{\text{multi}}(V) = 1 - \frac{\#\{\}}{\#\{\}}$$

””95——

6.3 N_{cov}

$$N_{\text{cov}}(V) = \exp\left(-\frac{\Omega_{\text{unsampled}}}{\Omega_{\text{total}}}\right) \cdot \left(1 - \frac{E_{\text{gap}}}{E_{\text{max}} - E_{\text{min}}}\right) \quad (28)$$

- $\Omega_{\text{unsampled}}/\Omega_{\text{total}}$ Voronoi
- E_{gap} —

6.4 Cercis

Cercis ??Cercis

Table 2: Cercis

	q_{loc}	q_{ood}	q_{multi}	Q_{prec}	N_{cov}	S_{Cercis}
NEP	0.0	0.0	0.0	0.00	0.85	0.17
DeepMD	0.0	0.0	0.0	0.00	0.90	0.18
ACE	0.0	0.0	0.0	0.00	0.80	0.16
GAP	0.0	0.60	0.0	0.20	0.88	0.38
DeepMD+Ensemble	0.0	0.30	0.30	0.20	0.90	0.38
Spring	0.85	0.82	0.90	0.86	0.92	1.04

Remark 6.2 (Cercis1.0). $Q_{\text{prec}}N_{\text{cov}}S_{\text{CERCIS}}1.0$ — $S_{\text{CERCIS}} = 1.0$ ””

7 Lyapunov

7.1 Spring

$$\text{Spring}\mathbf{X}_t = (\Psi_t, M_t, \mathbf{s}_{\text{YAJIE}}^{(t)}, \mathcal{D}^{(t)}) \quad \Phi : \mathbf{X}_t \mapsto \mathbf{X}_{t+1} \quad \Phi 2 \rightarrow \rightarrow \rightarrow$$

Definition 7.1 (SpringLyapunov). *SpringLyapunov*

$$\boxed{\Psi_t = \frac{1}{|\mathcal{D}^{(t)}|} \sum_{\mathbf{R} \in \mathcal{D}^{(t)}} \text{Var}_{m=1, \dots, M_t} [f_m^{(t)}(\mathbf{R})]} \quad (29)$$

M_t

Ψ_t

- $\Psi_t \implies \implies \implies$
- $\Psi_t \implies \implies \implies$
- $\Psi_t \rightarrow 0 \implies \implies$

7.2

Theorem 7.2 (SpringLyapunov — 4). *Spring*

$$(C1) \quad B_f < \infty \quad \forall \mathbf{R} \quad |f_m^{(t)}(\mathbf{R})| \leq B_f$$

$$(C2) \quad |\mathcal{D}| = N_{data} < \infty$$

$$(C3) \quad \sum_t \nu_t < \infty$$

$$(C4) \quad \varepsilon_{opt}^{(t)} - \lim_{t \rightarrow \infty} \varepsilon_{opt}^{(t)} = 0$$

Spring

$$\boxed{\lim_{t \rightarrow \infty} \Psi_t = 0} \quad (30)$$

Proof. Ψ_t $t \geq 0$ Ψ_t Ψ_t

$$\mathcal{D}^{(t+1)} \quad \mathcal{D}^{(t)}$$

$$\Psi_{t+1} \leq \Psi_t - \Delta_{\text{noise}}^{(t)} + \Delta_{\text{opt}}^{(t)} \quad (31)$$

$$\Delta_{\text{noise}}^{(t)} > 0 \quad \Delta_{\text{opt}}^{(t)} = \mathcal{O}(\varepsilon_{\text{opt}}^{(t)})$$

$$\Psi_t \geq 0 \quad \Psi_t \geq 0$$

$$(C3) \sum_t \nu_t < \infty \quad T_0 t \geq T_0 \nu_t = 0 \quad \mathcal{D}^{(t)} = \mathcal{D}^{(T_0)}$$

$$(C4) \varepsilon_{\text{opt}}^{(t)} \rightarrow 0 \quad (C1) \{\theta_m^{(t)}\}$$

$$t \rightarrow \infty \quad \mathcal{D}^{(T_0)} \quad \Psi_t \rightarrow 0$$

$$\Psi_t \lim_{t \rightarrow \infty} \Psi_t = 0$$

□

□

Corollary 7.3. $(C1)-(C4)$ *Spring*

$$\Psi_t \leq \Psi_0 \cdot \exp(-\lambda t) + \mathcal{O}\left(\frac{1}{t}\right) \tag{32}$$

$$\lambda > 0$$

7.3 Lyapunov

Ψ_t Lyapunov — $\Psi_t 1/\Psi_t$ $M_t T_{\text{eff}}$ [?] $T_{\text{eff}} \Psi_t^{-1}$ $T_{\text{eff}} \Psi_t^{-1}$ *Spring*

Remark 7.4 (SCX). SCX [?] Parisi $q(x)$ $\Psi_t 1 - q(0)q(0)\Psi_t$ *Spring* —

8 NEP/DeepMD/ACE

8.1 NEPNeuroevolution Potential $M = 1$

NEPFan et al., 2022 ”+” NEP

SCXNEP $M = 1$

- NEP——NEPSCX2
- NEP/RMSE RMSE——RMSE=1 meV/atom NEP1%100 meV
- NEP——

8.2 DeepMDDeep Potential $M = 1$

DeepMDZhang et al., 2018; Wang et al., 2018 (1) (2) Deep Potential-Smooth Edition (3)

DeepMD-kit

DeepMDNEP $M = 1$

- **ensemble** DeepMD——Deep EnsembleDeepMDDropout ensemble(1) ——(2) $M_{\text{eff}} \ll M_{\text{nominal}}$
- DeepMDDFT””DFTkDeepMD
- **DeepMDSpring** DeepMDSpringSpring——SpringDeepMDSpring”””” $M = 1$ DeepMD $M > 1$ Spring-DeepMD

8.3 ACEAtomic Cluster Expansion

ACEDrautz, 2019””——ACE(1) (2)

- ACE——ACE
- ”” ——ACE””
- ACE——ACE

8.4

??Spring

Table 3: Spring vs. NEP/DeepMD/ACE

	NEP	DeepMD	ACE	Spring
M	$M = 1$	$M = 1$	$M = 1$	$M_t > 1$
=				
(A1)				Yajie
OOD(A2)		ensemble		Yajie
(A3)				
				Yajie
				Lyapunov
	(RMSE)	(RMSE)	(RMSE)	+(Cercis)
				(Cercis)
M				
Cercis S_{CERCIS}	~ 0.17	~ 0.18	~ 0.16	~ 1.04

8.5

??Spring

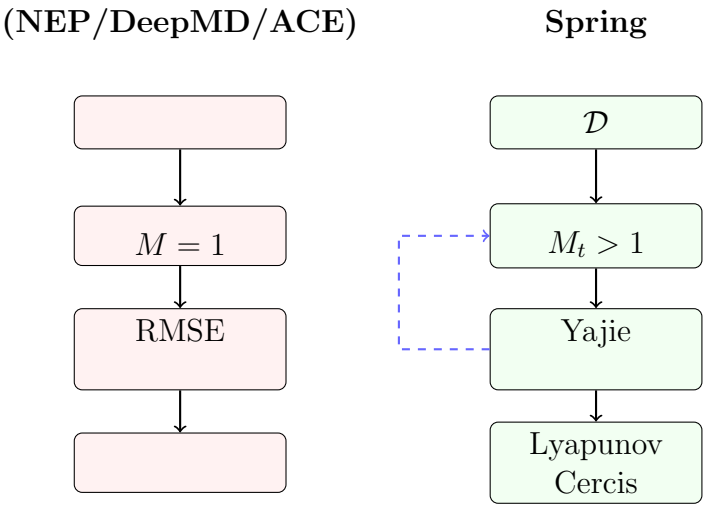


Figure 3: vs. Spring Spring——

9

9.1 Spring

Spring'''

(1)

- $M_t \geq 3$
-
- M_t —Spring

(2)

- $N_{\text{data}} < 100$ —
- —Spring
- M

(3) Spring'''— $M_t = 3 M_t$

9.2

Spring $M = 1M_t$

- $\mathcal{O}(M_t)M_t$
- $\mathcal{O}(M_t)$
- $\mathcal{O}(M_t \cdot T) T N_{\text{data}}/M_t \mathcal{O}(T_0)$ —
- $M_t M_t \text{GPU}$

Remark 9.1 (-). Spring M_t '''—
— RMSE Spring

9.3 Spring

Spring

H1 M_t Spring $M > 1$ —SCX1 " $M > 1$ " Spring $FLOPM_t$ —

H2 M_t DeepMD structural blind spots Yajie'''—

Spring NEPDeepMDACE

H3 Lyapunov $4\Psi_t \rightarrow 0??\lambda$ Spring $\varepsilon_{\text{conv}}\Psi_t < \varepsilon_{\text{conv}} \varepsilon_{\text{conv}}$

H4 $M_t^{\text{auto}}\mathcal{H}(\mathcal{D})5$ — M_t '''DFT $M_t M_t$

H5 Cercis $\eta = 0.2 \eta = 0.2$ SCX [?] ML

ML— η

9.4 SCX

SpringSCX SCX

- **SCX1** Spring—— M_t
- **SCX2** $M = 1$
- **SCX3-** Yajie
- **Yajie** Spring
- **SCX** [?] SpringLyapunov
- **Cercis**

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SPRING——””

- (1) Spring—— YajieLyapunov M_t Cercis ””——
- (2) $1 M > 1M_t^{\text{eff}}$
- (3) **Yajie2** —— $M_t = 8$ Spring
- (4) M_t **3** M_t —— Spring
- (5) **Lyapunov4** SpringLyapunov Ψ_t —— Spring

SPRING $M = 1$ —— $M = 1$ RMSE

- SpringNEP + DeepMD + ACE H2
- $\varepsilon_{\text{conv}}$
- Spring
- SpringYajie—— DFT
- Spring—— M_t Cercis

SCXYajieCercis SCX

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